

NASA SDS Product Specification

Level-2 Geocoded Unwrapped Interferogram

L2_GUNW

Rev E

JPL D-102272

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Revision	Cover Date	Sections Changed	ECR #	Reason, ECR Title, LRS #*
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Rev A (R2) Working Version 2	May 13, 2021	Sec 1.2, Sec 2.1, Sec 2.2, Sec 6	N/A	Merged info from Product Description Document (PDD). Removed reference to PDD. Added Section 6 on metadata cubes. LRR056764 for ISRO LRR056771 for NISAR Science Team
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Rev A (R3.1) Working Version	August 05, 2022	Cover. TBD, TBC tables, Acronym List	N/A	Requested fixes. LRR067551
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Rev A (R3.3)	June 6, 2023	All	N/A	Revised for R3.3 release URS CL# 23-3181

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Rev B (R3.4)	November 9, 2023	Cover page, Sec. 1.3, Sec. 2.2, Sec. 3.2.2; Sections 5.2, 5.3, 5.4, 5.5, 5.6	N/A	Modified date on cover page; Updated reference to RSLC product specification and NISAR products filename convention documents; Removed reference and link to EPDM; Corrected formation of blended pixel offsets layers in Sec. 2.2 and added reference to NISAR SDS ATBD; Modified Table 3.4 to reflect information in product XML specifications. Updated Sections 5.2, 5.3, 5.4, 5.5, 5.6.
Rev C (R4.0)	February 07, 2024	Cover page, Sec. 1.3, Figure 2-1, Table 2-1, Table 2-2, Sec. 2-2, Table 3-5; Sec. 3.7, Sec. 4.3, Sec. 4.4.2.2; Sec. 4.4.3, Appendix B	N/A	Added version number to cover page; Updated AD and AR; Updated Fig. 2-1; Revised RIFG, RUNW, GUNW description in Table 2-1; Revised product level description in Table 2-2; Clarified use of offset blending algorithm in Sec. 2.2; Removed unused attributes in Table 3-5; Updated product output grid features in Sec. 3.7; Revised and improved readability of Sec. 4.3; Added subsection on attitude state vectors; Improved description of geolocation grid Datasets in Sec. 4.4.3; Removed “Grid Alignment” paragraph from Appendix B. URS CL#24-1228.
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Rev E (R4.0.4)	November 8, 2024	Cover page, Sec. 5	N/A	Updated revision and date on cover page; Clarified in Section 5 format of timestamps and string data type. URS CL#24-6082
* *Include the JPL Limited Release System (LRS) clearance number for each revision to be shared with foreign partners.*				

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LIST OF TBC ITEMS

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1 INTRODUCTION

1.1 Purpose of Description

This document provides a specification of the NASA-ISRO Synthetic Aperture Radar (NISAR) L-SAR Level-2 (L2) Geocoded Unwrapped interferogram product to be generated by the NASA Science Data System (SDS) and provided to the Alaska Satellite Facility (ASF) Distributed Active Archive Center (DAAC). This data product is referenced by the short name GUNW.

1.2 Document Organization

Section 2 provides an overview of the product, including its purpose, and latency.

Section 3 provides the structure of the product, including granule definition, file organization, spatial resolution, temporal and spatial organization of the content.

Section 4 provides qualitative descriptions of the information provided in the product.

Section 5 provides a detailed identification of the individual fields within the GUNW product, including for example their units, size, and coordinates.

Section 6 provides a description of the metadata cube representation.

Appendix A provides a listing of the acronyms used in this document.

Appendix B provides a description of geolocation grids and projection systems used for the product.

1.3 Applicable and Reference Documents

Applicable documents levy requirements on areas addressed in this document. Reference documents are cited to provide additional information to readers. In case of conflict between the applicable documents and this document, the Project shall review the conflict to find the most effective resolution.

Applicable Documents

[AD1] NISAR NASA SDS Level 4 Requirements, JPL D-95655, Rev A, February 06, 2024

[AD2] NISAR NASA SDS Algorithm Development Plan, JPL D-95678, Initial, September 12, 2019

[AD3] NISAR Science Data Management and Archive Plan, JPL D-80828, June 1, 2016

[AD4] NISAR Science Management Plan, JPL D-76340, Rev A, August 14, 2018

[AD5] NISAR SDS ADT Calibration and Validation Plan, JPL D-102256, Rev A, November 20, 2023

[AD6] NISAR NASA SDS L4 Software Management Plan (SMP), JPL D-95656, Rev A, September 19, 2022

[AD7] ISO-19115-2, https://www.iso.org/obp/ui/#iso:std:iso:19115:-2:ed-2:v1:en

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Reference Documents

[RD1] NISAR NASA SDS Algorithm Theoretical Basis Document, JPL D-95677, Rev A, November 12, 2023

[RD2] EOSDIS Handbook, July 2016, retrieved from https://cdn.earthdata.nasa.gov/conduit/upload/5980/EOSDISHandbookWebFinaL2.pdf

[RD3] NISAR SDS L-SAR File Naming Conventions, JPL D-102255, Rev B, April 28, 2023

[RD4] NISAR L1_RSLC Product Specification Document, JPL D-102268, Rev C, February 07, 2024

[RD5] HDF5 documentation at https://portal.hdfgroup.org/display/HDF5/HDF5

[RD6] Eineder, M. (2003), Efficient simulation of SAR interferograms of large areas and of rugged terrain, IEEE Transactions on Geoscience and Remote Sensing, 41(6), 1415-1427

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2 PRODUCT OVERVIEW

2.1 Product Background

Each NASA SDS L0-L2 L-band product (Figure 2-1 and Table 2-1 Product dependency) is distributed as a single Hierarchical Data Format version 5 (HDF5, [RD5]) granule. All the metadata and imagery data are packaged in clearly defined sub-groups within the granule in compliance with the HDF5 specification [RD5]. The NISAR product level definitions are given in Table 2-2.

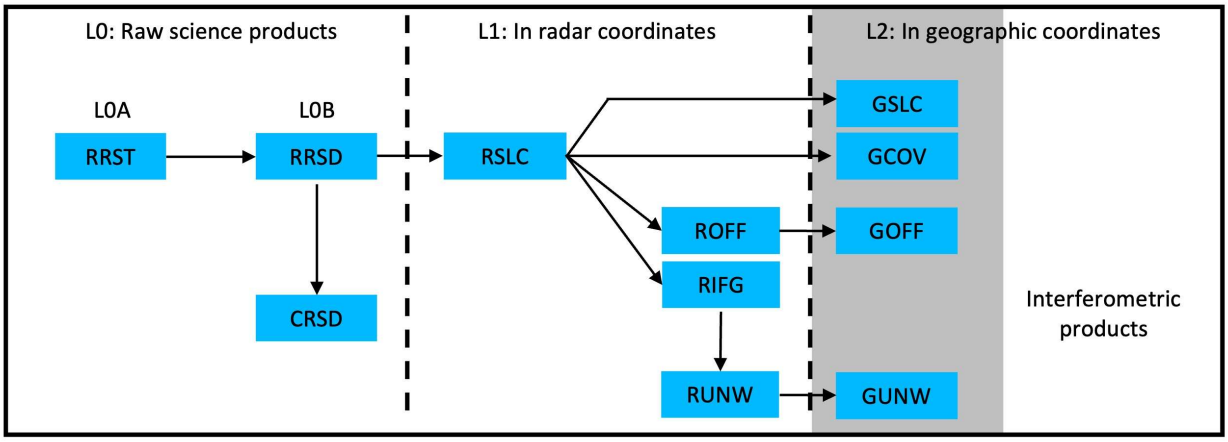


Figure 2-1 Product dependency.

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Table 2-1. Key to product dependency diagram.

Product L0	Scope	Description	Granule Size
Radar Raw Science Telemetry (RRST)	Global	This L0A product contains the raw downlinked data delivered to SDS	By downlinked files
Radar Raw Signal Data (RRSD)	Global	This L0B product is corrected, aligned radar pulse data derived from the RRST products and used for further processing	By radar observation, i.e., continuous data collected in a single radar mode
Calibration Raw Signal Data (CRSD)	Global	This L0B product contains instrument calibration data	By radar datatake, i.e., a sequence of observations for one radar-on period

Product L1	Scope	Description	Granule Size
Range-Doppler Single Look Complex (RSLC)	Global	The L1 RSLC product contains focused SAR images in range-Doppler coordinates. The RSLC is input to other L1 and L2 products	On pre-defined track/frame. High-resolution modes will have a high-res RSLC product and a background resolution RSLC product
Range-Doppler Nearest-Time Interferogram (RIFG)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only	Multi-looked interferogram in range-Doppler coordinates, ellipsoid and topographic phase flattened and formed with precise coregistration using geometrical offsets and high-resolution pixel offsets obtained from incoherent cross-correlation	On pre-defined track/frame
Range-Doppler Nearest-Time Pixel Offsets (ROFF)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only	Unfiltered and unculled layers of pixel offsets in range-Doppler coordinates with different resolutions obtained from incoherent cross-correlation	On pre-defined track/frame
Range-Doppler Nearest-Time Unwrapped Interferogram (RUNW)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only	Multi-looked unwrapped interferogram in range-Doppler coordinates, ellipsoid- and topography-flattened	On pre-defined track/frame

Product L2	Scope	Description	Granule Size
Geocoded SLC (GSLC)	Global and all channels	Single Look Complex SAR image on geocoded map coordinate system	On pre-defined track/frame
Geocoded Nearest-Time Pixel Offsets (GOFF)	Antarctica, Greenland, selected mountain glaciers. Nearest pair in time and co-pol channels only	Unfiltered and unculled layers of pixel offsets with different resolutions obtained from incoherent cross-correlation and geocoded on map coordinate system	On pre-defined track/frame
Geocoded Nearest-Time Unwrapped Interferogram (GUNW)	Global. Nearest pair in time and co-pol channels only	Geocoded, multi-looked, ellipsoid and topography flattened unwrapped interferogram	On pre-defined track/frame

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Product L2	Scope	Description	Granule Size
Geocoded Polarimetric Covariance Matrix (GCOV)	Global and all channels. Single/Dual/Quad pol	Geocoded, multi-looked polarimetric covariance matrix	On pre-defined track/frame

Table 2-2 NISAR product level descriptions defined by Science.

Product Level	Description
Level 0A	Unprocessed instrument data with all communications artifacts removed, but without reconstruction of missing data and sorting of samples from the instrument. May still contain bit errors and missing data that needs reconstruction
Level 0B	Reconstructed, time ordered, unprocessed instrument data at original resolution
Level 1	Processed instrument data, focused to full resolution complex images or derived radar parameters including interferometric phase and pixel offsets, in native radar coordinate system
Level 2	Focused radar imagery or derived radar parameters projected to a map coordinate system
Level 3	Derived geophysical parameters on a geocoded grids with the same or coarser posting as the Level 1 or Level 2 products

2.2 GUNW Product Overview

The GUNW product is a L2 product derived from the RIFG and RUNW products by geocoding the unwrapped phase and its associated data layers (i.e., coherence magnitude, ionospheric phase screen) on a geographical grid at 80 m posting. Geocoding is performed by using the orbit of the reference RSLC product and a Digital Elevation Model (DEM) to project the data onto a pre-defined Universal Transverse Mercator (UTM) or Polar stereographic projection system map grid (Appendix B: Geocoded Product Grids). The geocoding algorithm uses a bilinear interpolation for interpolating data layers with floating-point data types, Sinc for the complex wrapped interferogram, and nearest-neighbor interpolation for unsigned integer datasets (e.g., connected components mask).

The GUNW products also includes the wrapped complex interferogram (multilooked at 30 m in range-Doppler coordinates and geocoded at 20 m posting), the unwrapped interferometric phase in radians (80 m posting), the normalized interferometric coherence magnitude (20 m and 80 m posting), connected components mask, and sub-pixel offset layers obtained from incoherent cross-correlation. If an offset product in range-Doppler coordinates (e.g., ROFF) is available for the processed frame, the sub-pixel offset layers included in GUNW are obtained by optimally blending the multiresolution offset layers included in ROFF. The application of the offset layers

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blending is indicated by setting the Boolean flag in "/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/isOffsetsBlendingApplied" to "True". Conversely, if this Boolean flag is set to "False", the offset blending algorithm is not applied, and the sub-pixel offset layers included in GUNW are obtained by running incoherent cross-correlation on a coarse radar grid [RD1]. Regardless of the use of the offset blending algorithm, the pixel offsets layers are consistently geocoded at 80 m posting on the same geographical grid of the other data layers included in the GUNW product.

The GUNW product also includes an ionospheric phase screen layer and a layer quantifying its uncertainty. The ionospheric phase screen comes from the RUNW product and is estimated from the two spectral bands "frequencyA" and "frequencyB" whenever possible. In the case of mode transitions where continuity of spectral bands is impacted, a split spectrum ionospheric phase estimate is derived from the main imaging band ("frequencyA"). The estimated ionospheric phase screen is included as a layer in the product but not applied to the data layers within GUNW by default. The GUNW product also includes geocoded lookup tables for external phase corrections (e.g., solid Earth tides, hydrostatic and wet tropospheric phases). These phase corrections, when available, are not removed from the interferometric data.

The GUNW product with its group and basic properties is described in Section 4. The details of the data elements are given in Section 5. Metadata cubes are discussed in Section 6.

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3 PRODUCT ORGANIZATION

3.1 File Format

All NISAR standard products are in the HDF5 [RD5]. HDF5 is a general-purpose file format and programming library for storing scientific data. The National Center for Supercomputing Applications (NCSA) at the University of Illinois developed HDF to help scientists share data more easily. Use of the HDF library enables users to read HDF files regardless of the underlying computing environments. HDF files are equally accessible in Fortran, C/C++, and other high-level computation packages such as IDL, MATLAB or Python.

The HDF Group, a spin-off organization of the NCSA, is responsible for development and maintenance of HDF. Users should reference The HDF Group website at https://portal.hdfgroup.org/display/HDF5/HDF5 [RD5] to download HDF software and documentation.

HDF5 represents a significant departure from the conventions of previous versions of HDF. The changes that appear in HDF5 provide flexibility to overcome many of the limitations of previous releases. The basic building blocks have been largely redefined and are more powerful but less numerous. The key concepts of the HDF5 Abstract Data Model are Files, Groups, Datasets, Datatypes, Attributes, and Property Lists. The following sections provide a brief description of each of these key HDF5 concepts.

3.1.1 HDF5 File

A File is the abstract representation of a physical data file. Files are containers for HDF5 Objects. These Objects include Groups, Datasets, and named Datatypes.

3.1.2 HDF5 Group

Groups provide a means to organize the HDF5 Objects in HDF5 Files. Groups are containers for other Objects, including other Groups. In that sense, Groups are analogous to directories that are used to categorize and classify files in standard operating systems.

Groups and their nested objects can be accessed using a path-like notation, akin to the notation employed for accessing Unix directories. The root Group is “/”. A Group contained in root might be called “/myGroup”.

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3.1.3 HDF5 Dataset

The Dataset is the HDF5 component that stores user data. Each Dataset associates with a Dataspace that describes the data dimensions, as well as a Datatype that describes the basic unit of storage element. A Dataset can also have Attributes.

3.1.4 HDF5 Datatype

A Datatype describes a unit of data storage for Datasets and Attributes. Datatypes are subdivided into Atomic and Composite Types.

Atomic Datatypes are analogous to simple basic types in most programming languages. HDF5 Atomic Datatypes include Time, Bitfield, String, Reference, Opaque, Integer, and Float. Each atomic type has a specific set of properties. Examples of the properties associated with Atomic Datatypes are:

- Integers are assigned size, precision, offset, pad byte order, and are designated as signed or unsigned.
- Strings can be fixed or variable length, and may or may not be null-terminated.
- References are constructs within HDF5 Files that point to other HDF5 Objects in the same file.

HDF5 provides a large set of predefined Atomic Datatypes. Table 3-1 lists the Atomic Datatypes that are used in NISAR data products.

Table 3-1 HDF5 Atomic Datatypes.

HDF5 Atomic Datatypes	Description
H5T_STD_U8LE	unsigned, 8-bit, little-endian integer
H5T_STD_U16LE	unsigned, 16-bit, little-endian integer
H5T_STD_U32LE	unsigned, 32-bit, little-endian integer
H5T_STD_U64LE	unsigned, 64-bit, little-endian integer
H5T_STD_I8LE	signed, 8-bit, little-endian integer
H5T_STD_I16LE	signed, 16-bit, little-endian integer
H5T_STD_I32LE	signed, 32-bit, little-endian integer
H5T_STD_I64LE	signed, 64-bit, little-endian integer
H5T_IEEE_F32LE	32-bit, little-endian, IEEE floating point
H5T_IEEE_F64LE	64-bit, little-endian, IEEE floating point
H5T_C_S1	character string made up of one or more bytes

Derived Datatypes are user-defined variants of predefined Atomic Datatypes where the data organization has been modified at the bit-level. Derived data types are particularly useful for representing custom N-bit integers and floating-point numbers.

Composite Datatypes incorporate sets of Atomic Datatypes. Composite Datatypes include Array, Enumeration, Variable Length and Compound.

- The Array Datatype defines a multi-dimensional array that can be accessed atomically.

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- Variable Length presents a 1-D array element of variable length. Variable Length Datatypes are useful as building blocks of ragged arrays.
- Compound Datatypes are composed of named fields, each of which may be dissimilar Datatypes. Compound Datatypes are conceptually equivalent to structures in the C programming language.

Named Datatypes are explicitly stored as Objects within an HDF5 File. Named Datatypes provide a means to share Datatypes among Objects. Datatypes that are not explicitly stored as Named Datatypes are stored implicitly. They are stored separately for each Dataset or Attribute they describe.

The Derived and Compound Datatypes used in NISAR products are reported in Table 3-2.

Table 3-2 NISAR HDF5 Derived and Compound Datatypes.

Description	Comments
16-bit little-endian floating point	“binary16” half precision type in IEEE 754-2008 standard. Matches numpy.float16 type in Python. We will refer to this type as H5T_IEEE_F16LE or Float16 in our documents.

H5T_COMPOUND
{
16-bit little-endian floating-point "r";
16-bit little-endian floating-point "i";
}
	Complex numbers made up of two half precision floating point numbers.

H5T_COMPOUND
{
32-bit little-endian floating-point "r";
32-bit little-endian floating-point "i";
}
	Complex numbers made of two single precision floating point numbers.

H5T_COMPOUND
{
64-bit little-endian floating-point "r";
64-bit little-endian floating-point "i";
}
	Complex numbers made of two double precision floating point numbers.

3.1.5 HDF5 Attribute

An Attribute is a small aggregate of data that describes Groups or Datasets. Like Datasets, Attributes are also associated with a particular Dataspace and Datatype. Attributes cannot be subsetted or extended. Attributes themselves cannot have Attributes.

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3.2 NISAR File Organization

3.2.1 Groups

All NISAR HDF5 files are organized within a hierarchy of Groups, with no actual data at the root("/") level. Table 3-3 shows the general layout of the HDF5 files that are generated by the NISAR SDS.

Table 3-3 Group organization at the top level of a NISAR HDF5 File.

Group Name	Description
/science/LSAR/	All science data from the L-SAR instrument is organized under this group
/science/SSAR/	All science data from the S-SAR instrument is organized under this group
/science/[L/S]SAR/identification/	File level metadata for cataloging, archiving the granule

In the nominal baseline, L-SAR and S-SAR data will not appear in the same granule, even if they cover the same geographic area. Data structure described below the primary groups ("/science/LSAR/" for L-SAR and "/science/SSAR/" for S-SAR) will be the same for L-SAR and S-SAR products. The rest of the document from this point on describes the layout of the product containing L-SAR data. The specification for equivalent S-SAR data products will be the same except for the substitution of "LSAR" by "SSAR" in the dataset paths in the HDF5 granule.

3.2.2 File Level Metadata

Global metadata at the file level are currently given as Global Attributes shown in Table 3-4.

Metadata regarding the data in the particular granule are given in "/science/LSAR/identification/". These data are described further in Section 4.2 and Section 5.2.

Table 3-4 Global Attributes of GUNW.

Attribute	Format	Description	Value
Conventions	string	NetCDF-4 conventions adopted in this product	CF-1.7
Title	string	Product title	NISAR L2_GUNW Product
Institution	string	Name of producing agency	NASA JPL
mission_name	string	Mission name	NISAR
reference_document	string	Name and version of Product Description Document to use as reference for product	D-102272 NISAR NASA SDS Product Specification L2 Geocoded Unwrapped Interferogram
Contact	string	Contact information for producer of product	nisar-sds-ops@jpl.nasa.gov

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3.2.3 Variable Metadata - HDF5 Attributes

NISAR standards incorporate additional metadata that describe each HDF5 Dataset within the HDF5 file. Each of these metadata elements appear in an HDF5 Attribute that is directly associated with the HDF5 Dataset. Wherever possible, these HDF5 Attributes employ names that conform to the Climate and Forecast (CF) conventions.

Table 3-5 lists the CF names for the HDF5 Attributes that NISAR products typically employ.

Table 3-5 Common variable Attributes in HDF5 file.

Attribute	Description
_FillValue	The value used to represent missing or undefined data
description	Miscellaneous information about the data or the methods to generate it
long_name	A descriptive variable name that indicates its content
quality_flag	Names of variable quality flag(s) that are associated with this variable to indicate its quality
units	Unit of data
valid_max	Maximum theoretical value of the variable
valid_min	Minimum theoretical value of the variable

Some HDF5 datasets are populated with statistical Attributes. **Error! Reference source not found.**, Table 3-7 and Table 3-8 describe statistical Attributes added to real- and complex-valued, and mask HDF5 datasets, respectively. The list of real- and complex-valued and mask HDF5 datasets for the standard GUNW product is given in Table 3-9.

Table 3-6 Statistical Attributes for real-valued HDF5 Datasets.

Attribute	Description
min_value	Minimum value of a real-valued HDF5 Dataset
mean_value	Mean value of a real-valued HDF5 Dataset
max_value	Maximum value of a real-valued HDF5 Dataset
sample_stddev	Sample standard deviation of a real-valued HDF5 Dataset

Table 3-7. Statistical Attributes for complex-valued HDF5 Datasets.

Attribute	Description
min_real_value	Minimum value of the real part of a complex-valued HDF5 Dataset
mean_real_value	Mean value of the real part of a complex-valued HDF5 Dataset
max_real_value	Maximum value of the real part of a complex-valued HDF5 Dataset
sample_stddev_real	Sample standard deviation of the real part of a complex-valued HDF5 Dataset
min_imag_value	Minimum value of the imaginary part of a complex-valued HDF5 Dataset
mean_imag_value	Mean value of the imaginary part of a complex-valued HDF5 Dataset

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max_imag_value	Maximum value of the imaginary part of a complex-valued HDF5 Dataset
sample_stddev_imag	Sample standard deviation of the imaginary part of a complex-valued HDF5 Dataset

Table 3-8 Statistical Attributes for mask HDF5 Datasets.

Attribute	Description
layover_percentage	Percentage of pixels in layover
shadow_percentage	Percentage of pixels in shadow
layover_shadow_percentage	Percentage of pixels in layover and shadow
land_percentage	Percentage of pixels on land
water_percentage	Percentage of pixels on water bodies (e.g., ocean)

Table 3-9 GUNW HDF5 Datasets populated with statistical Attributes.

HDF5 Group	HDF5 Datasets	Dataset type
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/	mask	Four-valued
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/[HH/VV]/	unwrappedPhase, coherenceMagnitude, ionospherePhaseScreen	Real-valued
/science/LSAR/GUNW/grids/frequency/wrappedInterferogram/[HH/VV]/	coherenceMagnitude	Real-valued
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/[HH/VV]/	alongTrackOffset, slantRangeOffset	Real-valued
/science/LSAR/GUNW/metadata/radarGrid/	parallelBaseline, perpendicularBaseline	Real-Valued

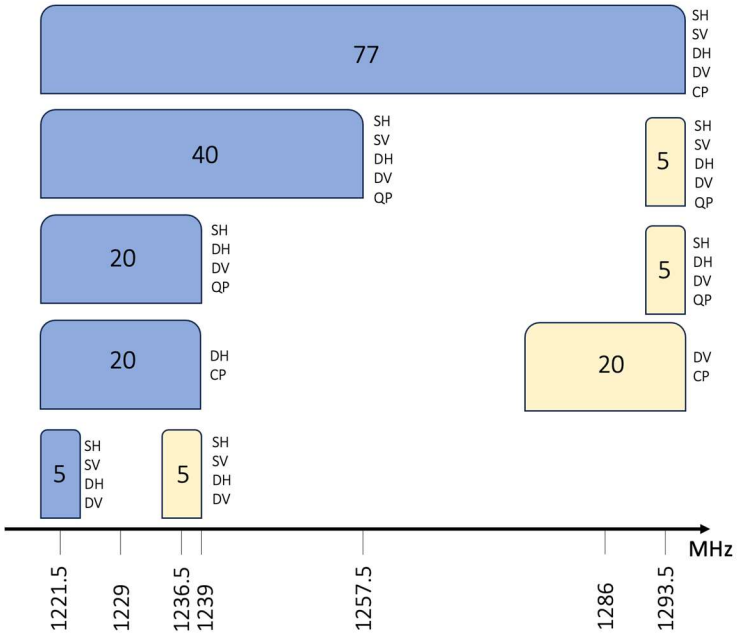
3.2.4 Variable Metadata - HDF5 Datasets

Users should note that the availability of specific polarizations (e.g., HH, VV) in any given NISAR GUNW product depends on the LSAR instrument's acquisition mode used over a specific region.

Figure 3-1 illustrates different possibilities of NISAR acquisitions in terms of range bandwidth and polarization of the transmitted and received signal. The LSAR instrument can acquire SAR data at the total 77 MHz range bandwidth or at parts of the spectrum as depicted in Figure 3-1. Acquisitions from the full 77 MHz spectrum and acquisitions from lower band of the spectrum (blue color in Figure 3-1) are stored in the "frequencyA" group of the NISAR L-band products (e.g., /science/LSAR/GUNW/grids/frequencyA). The SAR data at the higher band of the range spectrum (yellow color in Figure 3-1) correspond to "frequencyB". Under the current processing baseline, only data from the "frequencyA" Group is used for interferometric processing. Data in the "frequencyB" Group of reference and secondary RSLCs is not interferometrically processed and is therefore not included in GUNW products.

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The polarization content of the GUNW “frequencyA” Group is determined by the polarizations common to both reference and secondary RSLCs. For example, if the reference RSLC includes HH and HV polarizations while the secondary RSLC includes HH and VV polarizations, the common polarization would be HH, and only this polarization would be present in the GUNW product. Similarly, other polarization combinations may be derived based on the overlapping polarizations between the input RSLCs. Note, per current processing baseline, the cross-polarimetric channels HV and VH are not interferometrically processed.



Frequency A
Frequency B

SH: Single-Polarization H-transmit and H-receive (HH)
SV: Single-Polarization V-transmit and V-receive (VV)
DH: Dual-Polarization H-transmit and H- and V-receive (HH and HV)
DV: Dual-Polarization V-transmit and V- and H-receive (VV and VH)
QP: Quad-Polarization H- and V-transmit and H- and V-receive (HH,HV,VH,VV)
CP: Compact Polarization (RH, RV)

L-band mode Bandwidth [MHz]	Center frequency [MHz]	
	Frequency A	Frequency B
77	1257.5	NA
40+5	1239	1293.5
20+5	1229	1293.5
20+20	1229	1286
5+5	1221.5	1236.5

Figure 3-1 A visualization of possible NISAR L-band science acquisition modes

3.2.5 Georeferencing

NISAR L2 products contain georeferenced Datasets where the georeferencing information is provided in accordance with Climate and Forecast 1.7 (CF 1.7) conventions.

CF conventions require a "grid mapping" dataset which describes the coordinate system associated with the georeferenced Dataset. For NISAR L2 products, this grid mapping is represented by the Dataset "projection", which is included under the same Group as each georeferenced Dataset. Accordingly, each georeferenced Dataset contains an Attribute “grid_mapping”, whose value is always hard coded to the string "projection" (see Table 3-10).

The value of the "projection" Dataset is set to the European Petroleum Survey Group (EPSG) code of the associated georeferenced Dataset. The "projection" Dataset has Attributes with additional grid mapping information (see Table 3-10). More information about the projections used to represent NISAR L2 products is provided in Section 8.

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In addition to a grid mapping dataset, CF conventions use HDF5 Dimension Scales [RD5] to associate coordinates to each georeferenced Dataset. The Dimension Scales employed in NISAR L2 products are the "xCoordinates", "yCoordinates", and "heightAboveEllipsoid" Datasets, which represent the horizontal X- and Y-coordinates, and elevation Z-coordinates, respectively, and are located within the same Group as the associated georeferenced Dataset. These are one dimensional (1-D) vectors with lengths matching the associated Dataset's dimensions; each vector element corresponds to the grid-mapping location at the center of the georeferenced array pixels. "heightAboveEllipsoid" is only included for three-dimensional (3-D) georeferenced Datasets.

The complete listing of all georeferenced HDF5 Datasets within the GUNW product is given in Table 3-11. Note that the 3-D georeferenced Datasets are contained in the "radarGrid" Group; they are *metadata cubes* which represent the radar geometry in a compact form. Section 4.4.3 contains information about the “radarGrid” Group; Section 6 describes metadata cubes and their usage.

Table 3-10 Attributes of the HDF5 Dataset "projection" containing the grid mapping.

Attribute	Description
ellipsoid	Projection ellipsoid
epsg_code	Projection EPSG code
false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection
false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection
grid_mapping_name	Grid mapping variable name
inverse_flattening	Inverse flattening of the ellipsoidal figure
latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection
longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum
semi_major_axis	Semi-major axis
spatial_ref	Spatial reference
utm_zone_number	UTM zone number

Table 3-11 List of GUNW georeferenced Datasets.

HDF5 Group	HDF5 Datasets	Array Dimension
/science/LSAR/GUNW/grids/ frequencyA/wrappedInterferogram/[HH/VV]/	wrappedInterferogram coherenceMagnitude	2-D
/science/LSAR/GUNW/grids/ frequencyA/unwrappedInterferogram/[HH/VV]/	unwrappedPhase coherenceMagnitude connectedComponents ionospherePhaseScreen ionospherePhaseScreenUncertainty	2-D
/science/LSAR/GUNW/grids/ frequencyA/pixelOffsets/[HH/VV]/	alongTrackOffset slantRangeOffset correlationSurfacePeak	2-D
/science/LSAR/GUNW/metadata/radarGrid	slantRange	3-D

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	zeroDopplerAzimuthTime incidenceAngle, losUnitVectorX losUnitVectorY alongTrackUnitVectorX alongTrackUnitVectorY elevationAngle groundTrackVelocity	

3.3 Cloud Optimization

NISAR science data products utilize several special features of the HDF5 format to optimize file sizes and enable high-performance read access in a cloud environment. A key challenge of cloud data access is the latency associated with calls to the cloud storage Application Programming Interface (API), so the following strategies are used to minimize the number of cloud API calls needed per byte of data read:

- Chunks: Large datasets within the products use chunked storage. Every read operation thus fetches at least one entire chunk of data. The chunk size is nominally 512x512 pixels, though the precise chunk dimensions should be obtained using the H5Pget_chunk method of the HDF5 C API (or its equivalent in other language bindings).
- Compression: Data are written using a compression filter, minimizing the amount of data stored and hence transferred over the network. The HDF5 API handles decompression automatically.
- Paging: Files are created with the "paged" file space strategy (H5F_FSPACE_STRATEGY_PAGE in the HDF5 C API). These pages serve as the basic unit of allocation within the file. The page size is chosen larger than the chunk size so that both a chunk of data and its HDF5-internal metadata can be read in a single cloud API call. This parameter may be queried using the H5Pget_file_space_page_size method of the HDF5 C API.

Software that reads NISAR products stored on the cloud should take heed of the following recommendations:

- Set the page buffer size to a multiple of the file space page size using H5Pset_page_buffer_size in the HDF5 C API. This enables caching logic that reduces the number of cloud API calls in the file driver.
- Implement chunk-aligned data access patterns. Reads in multiples of the chunk size (and aligned with chunk boundaries) are most efficient.
- If other access patterns are desired, try setting the read cache large enough to hold all the chunks that may be re-read. For example, line-by-line access can still be efficient if the read cache is large enough to hold N lines, where N is the chunk dimension. That way lines can be read from the cache instead of fetching the same set of chunks N times over the network. The cache size may be set globally using the H5Pset_cache or locally with the H5Pset_chunk_cache methods of the HDF5 C API.

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Note that, in general, these optimizations require knowledge of the file contents. Therefore, the most robust approach is to open the file, inspect the contents (e.g., chunk size, page size, and dataset dimensions) and then re-open the file with optimal parameters.

3.4 Granule Definition

NISAR GUNW granules will conform to the Tiling Scheme being developed for the mission and are expected to have a ground footprint of 240 km x 240 km.

3.5 File Naming Convention

NISAR GUNW Granule names will conform to the Standard Product File Naming Scheme [RD3].

3.6 Temporal Organization

Temporal organization is not specifically applicable to the GUNW product, although it is generally arranged in order of increasing azimuth time.

3.7 Spatial Organization

The L2 data are arranged on a uniformly spaced, North-up and East-right grid, i.e., decreasing North or Y coordinate in the row direction and increasing East or X coordinate in the column direction following the row-major order convention of representing 2D raster arrays. Pixel-is-area convention is used to tag the raster layers with coordinate information.

3.8 Spatial Sampling and Resolution

Some salient features of the output grid for the GUNW product are:

1. The top-left corner of the top-left pixel will correspond to the same geographic coordinate for all imagery layers in an L-SAR GUNW product.
2. The main imaging band (“frequencyA”) is spatially averaged to the same posting, irrespective of the imaging. This allows for spatial mosaicking operations across instrument mode changes.

3.8.1 Mosaicking

The spatial sampling of the output grid has also been designed to facilitate along-track mosaicking of contiguous GUNW product granules if the user desires. See Section 8 for details on the common output grid used for all L2 products.

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3.8.2 Partially compressed SLC data

Partially compressed data in RSLC files will not be used to produce GUNW products.

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4 LEVEL 2 GEOCODED UNWRAPPED INTERFEROGRAM PRODUCT

In this section, we briefly describe the layout of GUNW data and associated metadata within the NISAR HDF5 file. Detailed description of Group and Dataset names can be found in Section 5. In this section, we focus on the organization of L-SAR instrument data within the file under the Group name "/science/LSAR/".

4.1 Dimensions and Shapes of Data

Information on the dimensions and shapes of the data items in various data tables is described as part of the metadata (Section 5.1). This information is useful both as part of the product identification and for setting up further processing, i.e., dimensioning arrays.

4.2 Product Identification

Information needed to identify this product is given under the Group "/science/LSAR/identification/" (Section 5.2). This includes information such as orbit, cycle, track, and frame numbers, acquisition times, a polygon representing the bounding box of the included imagery in geographic coordinates, product version, and product specification version (i.e., the version number of this document).

4.2.1 Composite Release Identifier

The Composite Release Identifier (CRID) is a global version identifier documenting the algorithms and the overall status of the science data system used to generate the product. The CRID follows the format *EPMMmp* where:

- **E (Environment)**: a single character representing the environment or the venue where the product was generated. It can assume the values:
  - *A*: if the product was generated in the Algorithm Development environment
  - *D*: if the product was generated in the Development environment
  - *P*: if the product was generated in the Production environment
  - *T*: if the product was generated in the Integration and Test (I&T) environment
- **P (Mission Phase)**: a single numerical digit indicating the mission phase in which the product was generated. It can assume the following values:
  - *0*: for pre-launch (Phase D)
  - *1*: for primary science phase operations (Phase E)
  - *2*: extended mission (Phase E)
  - *3*: post-operations (Phase F), decommissioning, end of mission processing

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- **MM (Major Release)**: two numeric digits monotonically increasing between 0 and 99. The Major Release resets to zero upon a change in the Mission Phase identifier. A change in the Major Release indicates a major change in the products i.e., a change to one or more algorithms or to the processing rules having a significant impact on the science content of the product. The Major Release stands as a composite of the versions of all the algorithms used in the science data production systems. Individual algorithm versions are allocated in the product metadata.
- **m (Minor Release):** a single numeric digit increasing monotonically between 0 and 1 indicating a minor update to the product and/or the data system. A change in the Minor Release identifier indicates minor algorithm changes (e.g., bug fixes, small functional updates) that do not have a significant impact on the product. The Minor Release identifier resets to zero upon every update to the Major Release identifier
- **P (Patch Release)**: a single numerical digit monotonically increasing between 0 and 1. A change in the Patch Release identifier indicates an update to the science data system software that has undergone the System Deployment Review to fix a critical bug. The Patch Release resets to zero upon updates to the Major Release or Minor Release identifiers.

4.3 Radar Imagery

The GUNW product's imagery layers and associated datasets are initially organized based on the center frequency within the Group "/science/LSAR/GUNW/grids/frequencyA/". Only the main NISAR imaging band ("frequencyA") will be processed for GUNW products. Imagery data is further categorized by their type. The Unwrapped Interferogram layer and associated Datasets are located under the Group "/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/". The Wrapped Interferogram and its associated datasets are situated under the Group "/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/". Subsequently, the cross-correlation sub-pixel offsets are located under the Group "/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/". Each of these Groups is further organized by polarization (TxRx), and by a final grouping. For example, the Unwrapped Phase Group could contain the Group "/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/". The imagery datasets reside within these polarization Groups. As an example, the Dataset "/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/unwrappedPhase" corresponds to the unwrapped phase Dataset derived from the "frequencyA" and "HH" polarization imagery layers within the reference and secondary input RSLCs.

The details of the data elements for the granule are given in Section 5.3.

4.4 Radar Metadata

The Group "/science/LSAR/GUNW/metadata/" includes a list of miscellaneous metadata needed to interpret the imagery (e.g., wrapped complex interferogram, unwrapped interferometric phase) included in the GUNW product.

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4.4.1 Processing Information

The Group "/science/LSAR/GUNW/metadata/processingInformation/" includes the processing parameters used to generate the GUNW product. This group also include a list of the used algorithms, and the inputs granules and files used to produce GUNW. For a complete description of this group, refer to Section 5.4.

4.4.1.1 Parameters

The Group "/science/LSAR/GUNW/metadata/processingInformation/parameters/" is further organized in seven Groups:

1. *common*: organized by frequency, and including the parameters derived by combining the information from the reference and secondary RSLC e.g., Doppler centroid and the Doppler bandwidth.
2. *reference*: including the reference terrain height of the reference RSLC and Boolean flags to indicate if the RSLC is the results of mixed mode processing and if RFI correction has been applied. This Group is further organized by frequency and includes some relevant parameters of the reference RSLC such as the slant range and zero Doppler time spacings, the slant range and the azimuth bandwidths, and the Doppler centroid.
3. *secondary*: this Group follows the same organization of *reference* but includes the corresponding metadata for the secondary RSLC.
4. *interferogram*: including the parameters used to generate the complex wrapped interferogram and the normalized interferometric correlation e.g., the common slant range and azimuth bandwidths and the number of looks in along-track and slant range directions used to generate the complex wrapped interferogram in radar coordinates.
5. *ionosphere*: including the parameters used to generate the ionosphere phase screen e.g., the bandwidth of the low and high sub-images used in the ionosphere phase estimation with the range split spectrum technique.
6. *pixelOffsets*: including the parameters (e.g., window size, search window, offset spacings) to generate the along-track and slant range layers of pixelOffsets in radar coordinates. This Group is further organized by frequency.
7. *geocoding*: including a set of Boolean flags indicating the corrections that have been applied while geocoding the pixel offsets layers from radar to geographical coordinates i.e., wet and dry troposphere correction, slant range and azimuth ionosphere corrections.

The Group *parameters* also contains the Dataset *runConfigurationContents* which includes a copy of the run configuration used for processing populated with all the processing options, parameter values, and input files.

4.4.1.2 Algorithms

The Group "/science/LSAR/GUNW/metadata/processingInformation/algorithms/" includes the name and the version of the software used to generate the product. The Group is further organized in distinct Groups identifying the processing steps used to generate the GUNW product:

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1. *coregistration*: including the algorithms used to perform the coarse and fine coregistration of the reference and secondary RSLCs (e.g., geometry coregistration, cross-correlation algorithm).
2. *interferogramFormation*: including the algorithms used to form the complex wrapped interferogram and the normalized interferometric correlation (e.g., flattening method)
3. *unwrapping*: including the algorithms used to perform phase unwrapping (e.g., unwrapping algorithm, unwrapping initializer, type of performed preprocessing of the wrapped interferometric phase).
4. *ionosphereEstimation*: including the algorithm used to perform the estimation of the ionosphere phase screen (e.g., outlier estimation and filling, unwrapping error correction).
5. *geocoding*: including the algorithms to geocode the different data layers contained in the GUNW product e.g., floating, integer, and complex geocoding interpolation and flags to identify which correction (i.e., slant range ionospheric delay) has been used during geocoding.

4.4.1.3 Input Files

The Group "/science/LSAR/GUNW/metadata/processingInformation/inputs/" includes the filenames of the input RSLC granules, configuration files, orbit files, and a description of the DEM used for processing used to generate the product.

4.4.2 Other Radar Metadata

4.4.2.1 Orbit

The reference RSLC orbit ephemeris used for generating the GUNW product is provided under "/science/LSAR/GUNW/metadata/orbit/" and further detailed in Section 5.5. This Group includes time-tagged antenna phase center position and velocity vectors in Earth Centered Earth Fixed (ECEF) Cartesian coordinates and information on the used orbit fidelity (e.g., Medium Orbit Ephemeris).

4.4.2.2 Attitude

The attitude state vectors of the reference RSLC used for generating the GUNW product can be found under the Group "/science/LSAR/GUNW/metadata/attitude/". This Group includes time-tagged quaternions and Euler angles representing the slant range plane from the antenna phase center in an ECEF Cartesian system.

4.4.3 Radar Grid

The group "/science/LSAR/GUNW/metadata/radarGrid/" contains information on the radar geometry of the reference RSLCs. The Datasets within this Group are given in the form of metadata cubes, referred to as *radar grid cubes*, that are organized over a three-dimensional geographic grid. The representation as data cubes, rather than two-dimensional rasters, is used to

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reduce the amount of space required to store radar geometry values within NISAR L2 products. This is possible because each radar grid cube contains slowly varying values in space that can be described by a low-resolution three-dimensional grid with sufficient accuracy.

These values, however, are usually required at the terrain height, often characterized by a fast-varying surface representing the local topography. A higher-resolution DEM can then be used to interpolate radar grid cubes and generate high-resolution maps of the corresponding radar geometry variable. Further information on metadata cubes along with examples on how to interpolate them using a reference DEM is provided in Section 6.

Radar grid cubes (for geocoded products) are provided in the same coordinate system as the product imagery with similar extents (bounding box) but coarser pixel spacing. The three-dimensional geographic grid is defined by the HDF5 Datasets "xCoordinates" (defining the east component), "yCoordinates" (north component), and "heightAboveEllipsoid" (height above the WGS84 ellipsoid), common to all radar grid cubes, and following the CF-1.7 convention.

The "radarGrid" Group also included the Datasets:

1. "referenceSlantRange" ("secondarySlantRange") and "referenceZeroDopplerAzimuthTime" ("secondaryZeroDopplerAzimuthTime") defining the zero-Doppler radar grid of the reference (secondary) RSLC. These Datasets contain respectively the range position in meters and the zero-Doppler azimuth time in seconds for each point of the geographical grid
2. "losUnitVectorX" and "losUnitVectorY" identifying the East and North components of the Line-Of-Sight (LOS) unit vector (i.e., the vector from the target to the sensor) in the East-North-Up (ENU) coordinate system for each point of the geographic grid. The Up component of the LOS unit vector can be simply derived from the East and North components as:

$$losUnitVectorZ = \sqrt{1 - losUnitVectorX^2 - losUnitVectorY^2}$$

3. "alongTrackUnitVectorX" and "alongTrackUnitVectorY" containing the East and North components of the along-track unit vector (i.e., the projection of the along-track vector at the ground height) in UTM coordinates
4. "incidenceAngle containing the incidence angle, i.e., the angle between the LOS vector and the normal to the ellipsoid at the target height
5. "elevationAngle" containing the elevation angle i.e., the angle between the LOS vector and the normal to the ellipsoid at the sensor
6. "groundTrackVelocity" containing the ground track velocity i.e., the absolute value of the platform velocity scaled at the target height
7. "slantRangeSolidEarthTidesPhase" and "alongTrackSolidEarthTidesPhase" representing the slant range and along-track phase components due to Solid Earth tides
8. "wetTroposphericPhaseScreen", "hydrostaticTroposphericPhaseScreen" representing the interferometric phase due to the wet and hydrostatic components of the tropospheric delay

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9. “perpendicularBaseline” and “parallelBaseline” containing the perpendicular and parallel component of the baseline between the reference and secondary RSLCs. The baseline components are only computed for the bottom and top heights of the radar grid cubes

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5 PRODUCT SPECIFICATION

The following subsections provide a detailed overview of the L2_GUNW product contents. In this context, the *string* data type refers to a fixed-length byte string, and all timestamps are represented in Coordinated Universal Time (UTC).

5.1 Dimensions and Shapes

To simplify the description of the layout of data within the HDF5 file, we will use a table of dimensions and shapes to represent the relationship between similarly sized Datasets. The entries in this table do not present actual Datasets in the HDF5. This table is meant to be a guide to interpreting the shapes of the datasets in subsequent subsections.

Table 5-1 Table of dimensions and shapes in GUNW product.

Name	Shape	Description
scalar	scalar	Scalar values
numberOfDatatakes	scalar	Number of datatakes in product
numberOfObservations	scalar	Number of observations in product
numberOfFrequencies	scalar	Number of L-SAR frequencies in product
numberOfFrequencyAPolarizations	scalar	Number of polarization layers associated with L-SAR frequencyA
wrappedInterferogramWidth	scalar	Number of pixels in all L-SAR datasets within the wrappedInterferogram group
wrappedInterferogramLength	scalar	Number of lines in all L-SAR datasets within the wrappedInterferogram group
complexWrappedInterferogramShape	(wrappedInterferogramLength, wrappedInterferogramWidth)	Shape associated with complex datasets in the wrappedInterferogram group
wrappedInterferogramShape	(wrappedInterferogramLength, wrappedInterferogramWidth)	Shape associated with real or integer datasets in the wrappedInterferogram group
unwrappedInterferogramWidth	scalar	Number of pixels in all L-SAR datasets within the unwrappedInterferogram group
unwrappedInterferogramLength	scalar	Number of lines in all L-SAR datasets within the unwrappedInterferogram group
unwrappedInterferogramShape	(unwrappedInterferogramLength, unwrappedInterferogramWidth)	Shape associated with real or integer datasets in the unwrappedInterferogram group
offsetShape	(offsetLength, offsetWidth)	Shape associated with Pixel Offset layers
offsetWidth	scalar	Number of pixels in Pixel Offset layers
offsetLength	scalar	Number of lines in all L-SAR frequencyA imagery datasets
radarCubeShape	(radarCubeHeight, radarCubeLength, radarCubeWidth)	Shape associated with metadata cubes
twoLayersCubeShape	(radarCubeWidth, radarCubeLength, twoLayersCubeHeight)	Shape associated with baseline metadata cubes
radarCubeHeight	scalar	Height dimension of the metadata cube
radarCubeLength	scalar	Length dimension of the metadata cube

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radarCubeWidth	scalar	Width dimension of the metadata cube
twoLayersCubeHeight	scalar	Height dimension of the baseline metadata cube
dopplerCentroidLength	scalar	Length dimension of Doppler centroid grid
dopplerCentroidWidth	scalar	Length dimension of Doppler centroid grid
dopplerCentroidShape	(dopplerCentroidLength, dopplerCentroidWidth)	Shape of the Doppler centroid grid
orbitListLength	scalar	Number of orbit state vectors
orbitShape	(orbitListLength, 3)	Shape of orbit state vector triplets dataset
attitudeListLength	scalar	Number of attitude state vectors
attitudeQuaternionShape	(attitudeListLength, 4)	Shape of attitude quaternion dataset
attitudeShape	(attitudeListLength, 3)	Shape of attitude Euler angle triplets dataset
numberOfInputL1Files	scalar	Number of input L1 granules
numberOfInputOrbitFiles	scalar	Number of input orbit files
numberOfInputConfigFiles	scalar	Number of input configuration files

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5.2 Product Identification

Table 5-2 NISAR HDF5 variables used for product identification.

Product identification variables	
/science/LSAR/identification/referenceAbsoluteOrbitNumber	
Type: UInt32	**Shape: scalar**
Description: Absolute orbit number for the reference RSLC	
/science/LSAR/identification/secondaryAbsoluteOrbitNumber	
Type: UInt32	**Shape: scalar**
Description: Absolute orbit number for the secondary RSLC	
/science/LSAR/identification/referenceIsJointObservation	
Type: string	**Shape: scalar**
Description: "True" if any portion of the reference RSLC was acquired in a joint observation mode (e.g., L-band and S-band simultaneously), "False" otherwise	
/science/LSAR/identification/secondaryIsJointObservation	
Type: string	**Shape: scalar**
Description: "True" if any portion of the secondary RSLC was acquired in a joint observation mode (e.g., L-band and S-band simultaneously), "False" otherwise	
/science/LSAR/identification/trackNumber	
Type: UInt32	**Shape: scalar**
Description: Track number	
/science/LSAR/identification/frameNumber	
Type: UInt16	**Shape: scalar**
Description: Frame number	
/science/LSAR/identification/missionId	
Type: string	**Shape: scalar**
Description: Mission identifier	
/science/LSAR/identification/processingCenter	
Type: string	**Shape: scalar**
Description: Data processing center	
/science/LSAR/identification/productType	
Type: string	**Shape: scalar**
Description: Product type	
/science/LSAR/identification/granuleId	
Type: string	**Shape: scalar**
Description: Unique granule identification name	
/science/LSAR/identification/productDoi	
Type: string	**Shape: scalar**
Description: Digital Object Identifier (DOI) for the product	
/science/LSAR/identification/productVersion	
Type: string	**Shape: scalar**
Description: Product version which represents the structure of the product and the science content governed by the algorithm, input data, and processing parameters	
/science/LSAR/identification/productSpecificationVersion	
Type: string	**Shape: scalar**
Description: Product specification version which represents the schema of this product	
/science/LSAR/identification/lookDirection	
Type: string	**Shape: scalar**

This document has been reviewed and determined not to contain export controlled technical data.

Description: Look direction, either "Left" or "Right"		
/science/LSAR/identification/orbitPassDirection		
Type: string	**Shape: scalar**	
Description: Orbit direction, either "Ascending" or "Descending"		
/science/LSAR/identification/referenceZeroDopplerStartTime		
Type: string	**Shape: scalar**	
Description: Azimuth start time (in UTC) of reference RSLC product in the format YYYY-mm-ddTHH:MM:SS.ssssssss		
/science/LSAR/identification/secondaryZeroDopplerStartTime		
Type: string	**Shape: scalar**	
Description: Azimuth start time (in UTC) of secondary RSLC product in the format YYYY-mm-ddTHH:MM:SS.ssssssss		
/science/LSAR/identification/referenceZeroDopplerEndTime		
Type: string	**Shape: scalar**	
Description: Azimuth stop time (in UTC) of reference RSLC product in the format YYYY-mm-ddTHH:MM:SS.ssssssss		
/science/LSAR/identification/secondaryZeroDopplerEndTime		
Type: string	**Shape: scalar**	
Description: Azimuth stop time (in UTC) of secondary RSLC product in the format YYYY-mm-ddTHH:MM:SS.ssssssss		
/science/LSAR/identification/plannedDatatakeId		
Type: string	**Shape: (numberOfDatatakes)**	
Description: List of planned datatakes included in the product		
/science/LSAR/identification/plannedObservationId		
Type: string	**Shape: (numberOfObservations)**	
Description: List of planned observations included in the product		
/science/LSAR/identification/isUrgentObservation		
Type: string	**Shape: scalar**	
Description: Flag indicating if observation is nominal ("False") or urgent ("True")		
/science/LSAR/identification/listOfFrequencies		
Type: string	**Shape: (numberOfFrequencies)**	
Description: List of frequency layers available in the product		
/science/LSAR/identification/diagnosticModeFlag		
Type: UByte	**Shape: scalar**	
Description: Indicates if the radar operation mode is a diagnostic mode (1-2) or DBFed science (0): 0, 1, or 2		
/science/LSAR/identification/productLevel		
Type: string	**Shape: scalar**	
Description: Product level. L0A: Unprocessed instrument data; L0B: Reformatted, unprocessed instrument data; L1: Processed instrument data in radar coordinates system; and L2: Processed instrument data in geocoded coordinates system		
/science/LSAR/identification/isGeocoded		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the product data is in the radar geometry ("False") or in the map geometry ("True")		
/science/LSAR/identification/boundingPolygon		
Type: string	**Shape: scalar**	
Description: OGR compatible WKT representing the bounding polygon of the image. Horizontal coordinates are WGS84 longitude followed by latitude (both in degrees), and the vertical coordinate is the height above the WGS84 ellipsoid in meters. The first point corresponds to the start-time, near-range radar coordinate, and the perimeter is traversed in counterclockwise order on the map. This means the traversal order in radar coordinates differs for left-looking and right-looking sensors. The polygon includes the four corners of the radar grid, with equal numbers of points distributed evenly in radar coordinates along each edge		
	ogr_geometry	polygon
	epsg	4326
/science/LSAR/identification/processingDateTime		
Type: string	**Shape: scalar**	
Description: Processing date and time (in UTC) in the format YYYY-mm-ddTHH:MM:SS		
/science/LSAR/identification/radarBand		
Type: string	**Shape: scalar**	
Description: Acquired frequency band, either "L" or "S"		

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/identification/platformName		
Type: string		**Shape: scalar**
Description: Name of the platform used to collect the remote sensing data provided in this product		
/science/LSAR/identification/instrumentName		
Type: string		**Shape: scalar**
Description: Name of the instrument used to collect the remote sensing data provided in this product		
/science/LSAR/identification/processingType		
Type: string		**Shape: scalar**
Description: Nominal (or) Urgent (or) Custom (or) Undefined		
/science/LSAR/identification/isDithered		
Type: string		**Shape: scalar**
Description: "True" if the pulse timing was varied (dithered) during acquisition, "False" otherwise		
/science/LSAR/identification/isMixedMode		
Type: string		**Shape: scalar**
Description: "True" if this product is generated from reference and secondary RSLCs with different range bandwidths, "False" otherwise		
/science/LSAR/identification/compositeReleaseId		
Type: string		**Shape: scalar**
Description: Unique version identifier of the science data production system		
/science/LSAR/identification/isFullFrame		
Type: string		**Shape: scalar**
Description: "True" if the product fully covers a NISAR frame, "False" if partial coverage		
	frameCoveragePercentage	Percentage of NISAR frame containing processed data
	thresholdPercentage	Threshold percentage used to determine if the product is full frame or partial frame

This document has been reviewed and determined not to contain export controlled technical data.

5.3 Radar Imagery

Table 5-3 NISAR HDF5 variables related to SAR imagery.

Product imagery variables		
/science/LSAR/GUNW/grids/frequencyA/listOfPolarizations		
Type: string		**Shape: (numberOfFrequencyAPolarizations)**
Description: List of processed polarization layers with frequency A		
/science/LSAR/GUNW/grids/frequencyA/centerFrequency		
Type: Float64		**Shape: scalar**
Description: Center frequency of the processed image in hertz		
	units	hertz
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name

This document has been reviewed and determined not to contain export controlled technical data.

	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/xCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/yCoordinates		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64		**Shape: (unwrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/xCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/yCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/xCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/yCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/mask		
Type: UByte		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Combination of water mask and a mask of subswaths of valid samples in the reference RSLC and geometrically-coregistered secondary RSLC. Each pixel value is a three-digit number: the most significant digit represents the water flag of that pixel the reference RSLC, where 1 is water and 0 is non-water; the second digit represents the subswath number of that pixel in the reference RSLC; the least significant digit represents the subswath number of that pixel in the secondary RSLC. A value of '0' in either subswath digit indicates an invalid sample in the corresponding RSLC		
	_FillValue	255
	percentage_water	Percentage of pixels over water bodies and ocean
	valid_min	0
	grid_mapping	projection
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.

This document has been reviewed and determined not to contain export controlled technical data.

	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name

This document has been reviewed and determined not to contain export controlled technical data.

	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/xCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/yCoordinates		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64		**Shape: (unwrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/xCoordinates		
Type: Float64		**Shape: (wrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/yCoordinates		
Type: Float64		**Shape: (wrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/xCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/yCoordinates		
Type: Float64		**Shape: (unwrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/xCoordinates		
Type: Float64		**Shape: (wrappedInterferogramWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/yCoordinates		
Type: Float64		**Shape: (wrappedInterferogramLength)**
Description: Y coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/unwrappedPhase		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Unwrapped interferogram between HH layers		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	radians
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/connectedComponents		
Type: UInt16		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**

This document has been reviewed and determined not to contain export controlled technical data.

Description: Connected components for HH layer		
	_FillValue	65535
	grid_mapping	projection
	units	1
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/coherenceMagnitude		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Coherence magnitude between HH layers		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	1
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/ionospherePhaseScreen		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Ionosphere phase screen		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	radians
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/HH/ionospherePhaseScreenUncertainty		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Uncertainty of the ionosphere phase screen		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	radians
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/unwrappedPhase		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Unwrapped interferogram between VV layers		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	radians
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/connectedComponents		
Type: UInt16		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Connected components for VV layer		
	_FillValue	65535
	grid_mapping	projection
	units	1
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/coherenceMagnitude		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Coherence magnitude between VV layers		

This document has been reviewed and determined not to contain export controlled technical data.

	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	1
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/ionospherePhaseScreen		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Ionosphere phase screen		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	radians
/science/LSAR/GUNW/grids/frequencyA/unwrappedInterferogram/VV/ionospherePhaseScreenUncertainty		
Type: Float32		**Shape: (unwrappedInterferogramLength, unwrappedInterferogramWidth)**
Description: Uncertainty of the ionosphere phase screen		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	radians
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/mask		
Type: UByte		**Shape: (wrappedInterferogramLength, wrappedInterferogramWidth)**
Description: Combination of water mask and a mask of subswaths of valid samples in the reference RSLC and geometrically-coregistered secondary RSLC. Each pixel value is a three-digit number: the most significant digit represents the water flag of that pixel the reference RSLC, where 1 is water and 0 is non-water; the second digit represents the subswath number of that pixel in the reference RSLC; the least significant digit represents the subswath number of that pixel in the secondary RSLC. A value of '0' in either subswath digit indicates an invalid sample in the corresponding RSLC		
	_FillValue	255
	grid_mapping	projection
	percentage_water	Percentage of pixels over water bodies and ocean
	valid_min	0
	valid_max	155
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/wrappedInterferogram		
Type: CFloat32		**Shape: (wrappedInterferogramLength, wrappedInterferogramWidth)**
Description: Complex wrapped interferogram between HH layers		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/HH/coherenceMagnitude		
Type: Float32		**Shape: (wrappedInterferogramLength, wrappedInterferogramWidth)**
Description: Coherence magnitude between HH layers		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points

This document has been reviewed and determined not to contain export controlled technical data.

	sample_stddev	Standard deviation of the numeric data points
	units	1
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/wrappedInterferogram		
Type: CFloat32		**Shape: (wrappedInterferogramLength, wrappedInterferogramWidth)**
Description: Complex wrapped interferogram between VV layers		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GUNW/grids/frequencyA/wrappedInterferogram/VV/coherenceMagnitude		
Type: Float32		**Shape: (wrappedInterferogramLength, wrappedInterferogramWidth)**
Description: Coherence magnitude between VV layers		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	1
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/pixelOffset/mask		
Type: UByte		**Shape: (offsetLength, offsetWidth)**
Description: Combination of water mask and a mask of subswaths of valid samples in the reference RSLC and geometrically-coregistered secondary RSLC. Each pixel value is a three-digit number: the most significant digit represents the water flag of that pixel the reference RSLC, where 1 is water and 0 is non-water; the second digit represents the subswath number of that pixel in the reference RSLC; the least significant digit represents the subswath number of that pixel in the secondary RSLC. A value of '0' in either subswath digit indicates an invalid sample in the corresponding RSLC		
	_FillValue	255
	grid_mapping	projection
	percentage_water	Percentage of pixels over water bodies and ocean
	valid_min	0
	valid_max	155
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/slantRangeOffset		
Type: Float32		**Shape: (offsetLength, offsetWidth)**
Description: Slant range offset		
	_FillValue	nan
	grid_mapping	projection

This document has been reviewed and determined not to contain export controlled technical data.

	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/alongTrackOffset		
Type: Float32		**Shape: (offsetLength, offsetWidth)**
Description: Along-track offset		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/correlationSurfacePeak		
Type: Float32		**Shape: (offsetLength, offsetWidth)**
Description: Normalized cross-correlation surface peak		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	1
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/slantRangeOffset		
Type: Float32		**Shape: (offsetLength, offsetWidth)**
Description: Slant range offset		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	meters

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/alongTrackOffset		
Type: Float32		**Shape: (offsetLength, offsetWidth)**
Description: Along-track offset		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/correlationSurfacePeak		
Type: Float32		**Shape: (offsetLength, offsetWidth)**
Description: Normalized correlation surface peak		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	units	1
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/xCoordinates		
Type: Float64		**Shape: (offsetWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/yCoordinates		
Type: Float64		**Shape: (offsetLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/xCoordinates		
Type: Float64		**Shape: (offsetWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/yCoordinates		
Type: Float64		**Shape: (offsetLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/HH/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing

This document has been reviewed and determined not to contain export controlled technical data.

	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GUNW/grids/frequencyA/pixelOffsets/VV/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters

This document has been reviewed and determined not to contain export controlled technical data.

5.4 Processing Information

Table 5-4 NISAR HDF5 variables related to processing parameters.

Processing-related variables	
/science/LSAR/GUNW/metadata/processingInformation/parameters/runConfigurationContents	
Type: string	**Shape: scalar**
Description: Contents of the run configuration file with parameters used for processing	
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/rfiMitigationApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if RFI mitigation has been applied to reference RSLC	
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/isMixedMode	
Type: string	**Shape: scalar**
Description: "True" if reference RSLC is a composite of data collected in multiple radar modes, "False" otherwise	
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/referenceTerrainHeight	
Type: Float32	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**
Description: Reference Terrain Height as a function of map coordinates for reference RSLC	
units	meters
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/slantRangeStart	
Type: Float64	**Shape: scalar**
Description: Slant range start distance for the reference RSLC	
units	meters
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/numberOfRangeSamples	
Type: UInt64	**Shape: scalar**
Description: Number of slant range samples for each azimuth line within the reference RSLC	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/numberOfAzimuthLines	
Type: UInt64	**Shape: scalar**
Description: Number of azimuth lines within the reference RSLC	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/slantRangeSpacing	
Type: Float64	**Shape: scalar**
Description: Slant range spacing of reference RSLC	
units	meters
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/zeroDopplerTimeSpacing	
Type: Float64	**Shape: scalar**
Description: Time interval in the along-track direction for reference RSLC raster layers	
units	seconds
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/zeroDopplerStartTime	
Type: string	**Shape: scalar**
Description: Azimuth start time (in UTC) of the reference RSLC product in the format YYYY-mm-ddTHH:MM:SS.sssssssss	
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/zeroDopplerEndTime	
Type: string	**Shape: scalar**
Description: Azimuth stop time (in UTC) of the reference RSLC product in the format YYYY-mm-ddTHH:MM:SS.sssssssss	
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/rangeBandwidth	
Type: Float64	**Shape: scalar**
Description: Processed slant range bandwidth for reference RSLC	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/azimuthBandwidth	

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64	**Shape: scalar**
Description: Processed azimuth bandwidth for reference RSLC	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/reference/frequencyA/dopplerCentroid	
Type: Float64	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**
Description: 2D LUT of Doppler centroid for frequency A	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/referenceTerrainHeight	
Type: Float32	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**
Description: Reference Terrain Height as a function of map coordinates for secondary RSLC	
units	meters
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/rfiMitigationApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if RFI mitigation has been applied to secondary RSLC	
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/isMixedMode	
Type: string	**Shape: scalar**
Description: "True" if secondary RSLC is a composite of data collected in multiple radar modes, "False" otherwise	
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/slantRangeStart	
Type: Float64	**Shape: scalar**
Description: Slant range start distance for the secondary RSLC	
units	meters
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/numberOfRangeSamples	
Type: UInt64	**Shape: scalar**
Description: Number of slant range samples for each azimuth line within the secondary RSLC	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/numberOfAzimuthLines	
Type: UInt64	**Shape: scalar**
Description: Number of azimuth lines within the secondary RSLC	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/slantRangeSpacing	
Type: Float64	**Shape: scalar**
Description: Slant range spacing of secondary RSLC	
units	meters
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/zeroDopplerTimeSpacing	
Type: Float64	**Shape: scalar**
Description: Time interval in the along-track direction for secondary RSLC raster layers	
units	seconds
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/zeroDopplerStartTime	
Type: string	**Shape: scalar**
Description: Azimuth start time (in UTC) of the secondary RSLC product in the format YYYY-mm-ddTHH:MM:SS.sssssssss	
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/zeroDopplerEndTime	
Type: string	**Shape: scalar**
Description: Azimuth stop time (in UTC) of the secondary RSLC product in the format YYYY-mm-ddTHH:MM:SS.sssssssss	
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/rangeBandwidth	
Type: Float64	**Shape: scalar**
Description: Processed slant range bandwidth for secondary RSLC	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/azimuthBandwidth	
Type: Float64	**Shape: scalar**
Description: Processed azimuth bandwidth for secondary RSLC	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/secondary/frequencyA/dopplerCentroid	
Type: Float64	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**

This document has been reviewed and determined not to contain export controlled technical data.

Description: 2D LUT of Doppler centroid for frequency A	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/common/frequencyA/dopplerCentroid	
Type: Float64	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**
Description: 2D LUT of common Doppler centroid between reference and secondary RSLCs	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/common/frequencyA/dopplerBandwidth	
Type: Float64	**Shape: scalar**
Description: Common Doppler bandwidth between reference and secondary RSLCs	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/rangeBandwidth	
Type: Float64	**Shape: scalar**
Description: Processed slant range bandwidth for frequency A interferometric layers	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/azimuthBandwidth	
Type: Float64	**Shape: scalar**
Description: Processed azimuth bandwidth for frequency A interferometric layers	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/commonBandRangeFilterApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if common band range filter has been applied	
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/commonBandAzimuthFilterApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if common band azimuth filter has been applied	
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/numberOfRangeLooks	
Type: UInt32	**Shape: scalar**
Description: Number of looks applied in the slant range direction to form the wrapped interferogram	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/numberOfAzimuthLooks	
Type: UInt32	**Shape: scalar**
Description: Number of looks applied in the along-track direction to form the wrapped interferogram	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/ellipsoidalFlatteningApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the interferometric phase has been flattened with respect to a zero height ellipsoid	
/science/LSAR/GUNW/metadata/processingInformation/parameters/wrappedInterferogram/frequencyA/topographicFlatteningApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the interferometric phase has been flattened with respect to topographic height using a DEM	
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/rangeBandwidth	
Type: Float64	**Shape: scalar**
Description: Processed slant range bandwidth for frequency A interferometric layers	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/azimuthBandwidth	

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64	**Shape: scalar**
Description: Processed azimuth bandwidth for frequency A interferometric layers	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/commonBandRangeFilterApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if common band range filter has been applied	
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/commonBandAzimuthFilterApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if common band azimuth filter has been applied	
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/numberOfRangeLooks	
Type: UInt32	**Shape: scalar**
Description: Number of looks applied in the slant range direction to form the unwrapped interferogram	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/numberOfAzimuthLooks	
Type: UInt32	**Shape: scalar**
Description: Number of looks applied in the along-track direction to form the unwrapped interferogram	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/ellipsoidalFlatteningApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the interferometric phase has been flattened with respect to a zero height ellipsoid	
/science/LSAR/GUNW/metadata/processingInformation/parameters/unwrappedInterferogram/frequencyA/topographicFlatteningApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the interferometric phase has been flattened with respect to topographic height using a DEM	
/science/LSAR/GUNW/metadata/processingInformation/parameters/ionosphere/lowBandBandwidth	
Type: Float32	**Shape: scalar**
Description: Slant range bandwidth of the low sub-band image	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/ionosphere/highBandBandwidth	
Type: Float32	**Shape: scalar**
Description: Slant range bandwidth of the high sub-band image	
units	hertz
/science/LSAR/GUNW/metadata/processingInformation/parameters/geocoding/rangeIonosphericCorrectionApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the range ionospheric correction is applied to improve geolocation	
/science/LSAR/GUNW/metadata/processingInformation/parameters/geocoding/azimuthIonosphericCorrectionApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the azimuth ionospheric correction is applied to improve geolocation	
/science/LSAR/GUNW/metadata/processingInformation/parameters/geocoding/hydrostaticTroposphericCorrectionApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the hydrostatic tropospheric correction is applied to improve geolocation	
/science/LSAR/GUNW/metadata/processingInformation/parameters/geocoding/wetTroposphericCorrectionApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if the wet tropospheric correction is applied to improve geolocation	
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/alongTrackWindowSize	
Type: UInt32	**Shape: scalar**
Description: Along-track cross-correlation window size in pixels	

This document has been reviewed and determined not to contain export controlled technical data.

units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/slantRangeWindowSize	
Type: UInt32	**Shape: scalar**
Description: Slant range cross-correlation window size in pixels	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/alongTrackSearchWindowSize	
Type: UInt32	**Shape: scalar**
Description: Along-track cross-correlation search window size in pixels	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/slantRangeSearchWindowSize	
Type: UInt32	**Shape: scalar**
Description: Slant range cross-correlation search window size in pixels	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/alongTrackSkipWindowSize	
Type: UInt32	**Shape: scalar**
Description: Along-track cross-correlation skip window size in pixels	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/slantRangeSkipWindowSize	
Type: UInt32	**Shape: scalar**
Description: Slant range cross-correlation skip window size in pixels	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/correlationSurfaceOversampling	
Type: UInt32	**Shape: scalar**
Description: Oversampling factor of the cross-correlation surface	
units	1
/science/LSAR/GUNW/metadata/processingInformation/parameters/pixelOffsets/frequencyA/isOffsetsBlendingApplied	
Type: string	**Shape: scalar**
Description: Flag to indicate if pixel offsets are the results of blending multi-resolution layers of pixel offsets	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/softwareVersion	
Type: string	**Shape: scalar**
Description: Software version used for processing	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/coregistrationMethod	
Type: string	**Shape: scalar**
Description: RSLC coregistration method	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/geometryCoregistration	
Type: string	**Shape: scalar**
Description: Geometry coregistration algorithm	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/crossCorrelation	
Type: string	**Shape: scalar**
Description: Cross-correlation algorithm for sub-pixel offsets computation	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/resampling	
Type: string	**Shape: scalar**
Description: Secondary RSLC resampling algorithm	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/crossCorrelationOutliers	
Type: string	**Shape: scalar**
Description: Outliers identification algorithm	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/crossCorrelationFilling	

This document has been reviewed and determined not to contain export controlled technical data.

Type: string	**Shape: scalar**
Description: Outliers data filling algorithm for cross-correlation offsets	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/coregistration/crossCorrelationFilterKernel	
Type: string	**Shape: scalar**
Description: Filtering algorithm for cross-correlation offsets	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/interferogramFormation/multilooking	
Type: string	**Shape: scalar**
Description: Multilooking algorithm	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/interferogramFormation/wrappedInterferogramFiltering	
Type: string	**Shape: scalar**
Description: Algorithm used to filter the wrapped interferogram prior to phase unwrapping	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/interferogramFormation/flatteningMethod	
Type: string	**Shape: scalar**
Description: Algorithm used to flatten the wrapped interferogram	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/unwrapping/unwrappingAlgorithm	
Type: string	**Shape: scalar**
Description: Algorithm used for phase unwrapping	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/unwrapping/unwrappingInitializer	
Type: string	**Shape: scalar**
Description: Algorithm used to initialize phase unwrapping	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/unwrapping/costMode	
Type: string	**Shape: scalar**
Description: Cost mode algorithm for phase unwrapping	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/unwrapping/preprocessing/wrappedPhaseOutliers	
Type: string	**Shape: scalar**
Description: Algorithm identifying outliers in the wrapped interferogram	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/unwrapping/preprocessing/wrappedPhaseFilling	
Type: string	**Shape: scalar**
Description: Outliers data filling algorithm for phase unwrapping preprocessing	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/ionosphereEstimation/ionosphereAlgorithm	
Type: string	**Shape: scalar**
Description: Algorithm used to estimate ionosphere phase screen	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/ionosphereEstimation/ionosphereOutliers	
Type: string	**Shape: scalar**
Description: Algorithm identifying outliers in unfiltered ionosphere phase screen	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/ionosphereEstimation/ionosphereFilling	
Type: string	**Shape: scalar**
Description: Outliers data filling algorithm for ionosphere phase estimation	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/ionosphereEstimation/ionosphereFiltering	
Type: string	**Shape: scalar**
Description: Filtering algorithm for ionosphere phase screen computation	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/ionosphereEstimation/unwrappingErrorCorrection	
Type: string	**Shape: scalar**
Description: Algorithm correcting unwrapping errors in sub-band unwrapped interferograms	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/geocoding/demInterpolation	
Type: string	**Shape: scalar**
Description: DEM interpolation algorithm	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/geocoding/floatingGeocodingInterpolation	
Type: string	**Shape: scalar**
Description: Geocoding interpolation algorithm for floating point datasets	

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/science/LSAR/GUNW/metadata/processingInformation/algorithms/geocoding/integerGeocodingInterpolation	
Type: string	**Shape: scalar**
Description: Geocoding interpolation algorithm for integer datasets	
/science/LSAR/GUNW/metadata/processingInformation/algorithms/geocoding/complexGeocodingInterpolation	
Type: string	**Shape: scalar**
Description: Geocoding interpolation algorithm for complex-valued datasets	
/science/LSAR/GUNW/metadata/processingInformation/inputs/l1ReferenceSlcGranules	
Type: string	**Shape: (numberOfInputL1Files)**
Description: List of input reference L1 RSLC products used	
/science/LSAR/GUNW/metadata/processingInformation/inputs/l1SecondarySlcGranules	
Type: string	**Shape: (numberOfInputL1Files)**
Description: List of input secondary L1 RSLC products used	
/science/LSAR/GUNW/metadata/processingInformation/inputs/orbitFiles	
Type: string	**Shape: (numberOfInputOrbitFiles)**
Description: List of input orbit files used	
/science/LSAR/GUNW/metadata/processingInformation/inputs/configFiles	
Type: string	**Shape: (numberOfInputConfigFiles)**
Description: List of input config files used	
/science/LSAR/GUNW/metadata/processingInformation/inputs/demSource	
Type: string	**Shape: scalar**
Description: Description of the input digital elevation model (DEM)	

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5.5 Other Radar Metadata

Table 5-5 NISAR HDF5 variables related to useful radar metadata.

Radar metadata-related variables		
/science/LSAR/GUNW/metadata/orbit/temporalBaseline		
Type: UInt16	**Shape: scalar**	
Description: Time interval between reference and secondary RSLCs		
	units	days
/science/LSAR/GUNW/metadata/orbit/reference/interpMethod		
Type: string	**Shape: scalar**	
Description: Orbit interpolation method, either "Hermite" or "Legendre"		
/science/LSAR/GUNW/metadata/orbit/reference/time		
Type: Float64	**Shape: (orbitListLength)**	
Description: Time vector record. This record contains the time since UTC epoch corresponding to position and velocity records		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GUNW/metadata/orbit/reference/position		
Type: Float64	**Shape: (orbitListLength, tripletxyz)**	
Description: Position vector record. This record contains the platform position data with respect to WGS84 G1762 reference frame		
	units	meters
/science/LSAR/GUNW/metadata/orbit/reference/velocity		
Type: Float64	**Shape: (orbitListLength, tripletxyz)**	
Description: Velocity vector record. This record contains the platform velocity data with respect to WGS84 G1762 reference frame		
	units	meters / second
/science/LSAR/GUNW/metadata/orbit/reference/orbitType		
Type: string	**Shape: scalar**	
Description: Orbit product type, either "FOE", "NOE", "MOE", "POE", or "Custom", where "FOE" stands for Forecast Orbit Ephemeris, "NOE" is Near real-time Orbit Ephemeris, "MOE" is Medium precision Orbit Ephemeris, and "POE" is Precise Orbit Ephemeris		
/science/LSAR/GUNW/metadata/orbit/secondary/interpMethod		
Type: string	**Shape: scalar**	
Description: Orbit interpolation method, either "Hermite" or "Legendre"		
/science/LSAR/GUNW/metadata/orbit/secondary/time		
Type: Float64	**Shape: (orbitListLength)**	
Description: Time vector record. This record contains the time since UTC epoch corresponding to position and velocity records		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GUNW/metadata/orbit/secondary/position		
Type: Float64	**Shape: (orbitListLength, tripletxyz)**	
Description: Position vector record. This record contains the platform position data with respect to WGS84 G1762 reference frame		
	units	meters
/science/LSAR/GUNW/metadata/orbit/secondary/velocity		
Type: Float64	**Shape: (orbitListLength, tripletxyz)**	
Description: Velocity vector record. This record contains the platform velocity data with respect to WGS84 G1762 reference frame		
	units	meters / second
/science/LSAR/GUNW/metadata/orbit/secondary/orbitType		
Type: string	**Shape: scalar**	
Description: Orbit product type, either "FOE", "NOE", "MOE", "POE", or "Custom", where "FOE" stands for Forecast Orbit Ephemeris, "NOE" is Near real-time Orbit Ephemeris, "MOE" is Medium precision Orbit Ephemeris, and "POE" is Precise Orbit Ephemeris		
/science/LSAR/GUNW/metadata/attitude/reference/time		
Type: Float64	**Shape: (attitudeListLength)**	

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Description: Time vector record. This record contains the time since UTC epoch corresponding to attitude and quaternion records		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GUNW/metadata/attitude/reference/quaternions		
Type: Float64		**Shape: (attitudeListLength, quaternions)**
Description: Attitude quaternions (q0, q1, q2, q3)		
	units	1
/science/LSAR/GUNW/metadata/attitude/reference/eulerAngles		
Type: Float64		**Shape: (attitudeListLength, tripletxyz)**
Description: Attitude Euler angles (roll, pitch, yaw)		
	units	degrees
/science/LSAR/GUNW/metadata/attitude/reference/attitudeType		
Type: string		**Shape: scalar**
Description: Attitude type, either "FRP", "NRP", "PRP, or "Custom", where "FRP" stands for Forecast Radar Pointing, "NRP" is Near Real-time Pointing, and "PRP" is Precise Radar Pointing		
/science/LSAR/GUNW/metadata/attitude/secondary/time		
Type: Float64		**Shape: (attitudeListLength)**
Description: Time vector record. This record contains the time since UTC epoch corresponding to attitude and quaternion records		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GUNW/metadata/attitude/secondary/quaternions		
Type: Float64		**Shape: (attitudeListLength, quaternions)**
Description: Attitude quaternions (q0, q1, q2, q3)		
	units	1
/science/LSAR/GUNW/metadata/attitude/secondary/eulerAngles		
Type: Float64		**Shape: (attitudeListLength, tripletxyz)**
Description: Attitude Euler angles (roll, pitch, yaw)		
	units	degrees
/science/LSAR/GUNW/metadata/attitude/secondary/attitudeType		
Type: string		**Shape: scalar**
Description: Attitude type, either "FRP", "NRP", "PRP, or "Custom", where "FRP" stands for Forecast Radar Pointing, "NRP" is Near Real-time Pointing, and "PRP" is Precise Radar Pointing		

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5.6 Radar Grid

Table 5-6 NISAR HDF5 variables related to metadata cube.

Metadata cube-related variables		
/science/LSAR/GUNW/metadata/radarGrid/referenceSlantRange		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Slant range of the reference RSLC in meters		
	_FillValue	nan
	grid_mapping	projection
	long_name	Slant range
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/hydrostaticTroposphericPhaseScreen		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Hydrostatic component of the troposphere phase screen		
	_FillValue	nan
	grid_mapping	projection
	units	radians
/science/LSAR/GUNW/metadata/radarGrid/wetTroposphericPhaseScreen		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Wet component of the troposphere phase screen		
	_FillValue	nan
	grid_mapping	projection
	units	radians
/science/LSAR/GUNW/metadata/radarGrid/slantRangeSolidEarthTidesPhase		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Solid Earth tides phase along slant range direction		
	_FillValue	nan
	grid_mapping	projection
	units	radians
/science/LSAR/GUNW/metadata/radarGrid/referenceZeroDopplerAzimuthTime		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Zero Doppler azimuth time in seconds since UTC epoch of the reference RSLC image		
	_FillValue	nan
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GUNW/metadata/radarGrid/incidenceAngle		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Incidence angle is defined as the angle between the LOS vector and the normal to the ellipsoid at the target height		
	valid_max	90.0
	valid_min	0.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Incidence angle
	units	degrees
/science/LSAR/GUNW/metadata/radarGrid/losUnitVectorX		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: East component of unit vector of LOS from target to sensor		
	valid_max	1.0
	valid_min	-1.0

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	_FillValue	nan
	grid_mapping	projection
	long_name	LOS unit vector X
	units	1
/science/LSAR/GUNW/metadata/radarGrid/losUnitVectorY		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: North component of unit vector of LOS from target to sensor		
	valid_max	1.0
	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	LOS unit vector Y
	units	1
/science/LSAR/GUNW/metadata/radarGrid/alongTrackUnitVectorX		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: East component of unit vector along ground track		
	valid_max	1.0
	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Along-track unit vector X
	units	1
/science/LSAR/GUNW/metadata/radarGrid/alongTrackUnitVectorY		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: North component of unit vector along ground track		
	valid_max	1.0
	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Along-track unit vector Y
	units	1
/science/LSAR/GUNW/metadata/radarGrid/elevationAngle		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Elevation angle is defined as the angle between the LOS vector and the normal to the ellipsoid at the sensor		
	valid_max	90.0
	valid_min	0.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Elevation angle
	units	degrees
/science/LSAR/GUNW/metadata/radarGrid/groundTrackVelocity		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Absolute value of the platform velocity scaled at the target height		
	_FillValue	nan
	grid_mapping	projection
	long_name	Ground track velocity
	units	meters / second
/science/LSAR/GUNW/metadata/radarGrid/secondaryZeroDopplerAzimuthTime		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Zero Doppler azimuth time in seconds since UTC epoch of the secondary RSLC image		
	_FillValue	nan
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GUNW/metadata/radarGrid/secondarySlantRange		

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Type: Float64		**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**
Description: Slant range of the secondary RSLC in meters		
	_FillValue	nan
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/parallelBaseline		
Type: Float32		**Shape: (radarCubeWidth, radarCubeLength, twoLayersCubeHeight)**
Description: Parallel component of the InSAR baseline		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	long_name	Parallel baseline
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/perpendicularBaseline		
Type: Float32		**Shape: (radarCubeWidth, radarCubeLength, twoLayersCubeHeight)**
Description: Perpendicular component of the InSAR baseline		
	_FillValue	nan
	grid_mapping	projection
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	long_name	Perpendicular baseline
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/xCoordinates		
Type: Float64		**Shape: (radarCubeWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/yCoordinates		
Type: Float64		**Shape: (radarCubeWidth)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/heightAboveEllipsoid		
Type: Float64		**Shape: (radarCubeHeight)**
Description: Height values above WGS84 Ellipsoid corresponding to the radar grid		
	standard_name	height_above_reference_ellipsoid
	units	meters
/science/LSAR/GUNW/metadata/radarGrid/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name

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	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number

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6 METADATA CUBE

In this section, we provide an overview of the metadata cubes used to store spatially-varying ancillary data in the secondary layers of the NISAR L-SAR product HDF5 granules. Note that this sparse representation is to assist users in ingesting and analyzing NISAR products within existing GIS software and is not meant to replace traditional representations of SAR data within the product granules or traditional processing approaches with radar geometry-aware software.

Metadata cubes are represented as three-dimensional arrays in the NISAR product HDF5 modules (Figure 6-1). The axes of the array are interpreted as (height, increasing azimuth time, and increasing slant range) in case of radar geometry products and as (height, decreasing northing, and increasing easting) in case of geocoded products. The data is organized with height as the first axis, as this allows one to directly ingest data as GCPs or rasters into existing GIS software. Each height layer is the same size. Metadata cubes will have fixed grid spacing (3 km in azimuth/northing x 1 km in slant range/easting x 1.5 km in height and will allow for easy merging when multiple products along the same imaging track are to be concatenated. The metadata fields on this coarse resolution grid will be evaluated using traditional radar processing approaches without approximations. The metadata cube will also span a field slightly larger than the original image product to allow users to interpolate data without introducing edge effects. Such low-resolution representation of slowly varying parameters has been demonstrated for InSAR products and processing [RD6].

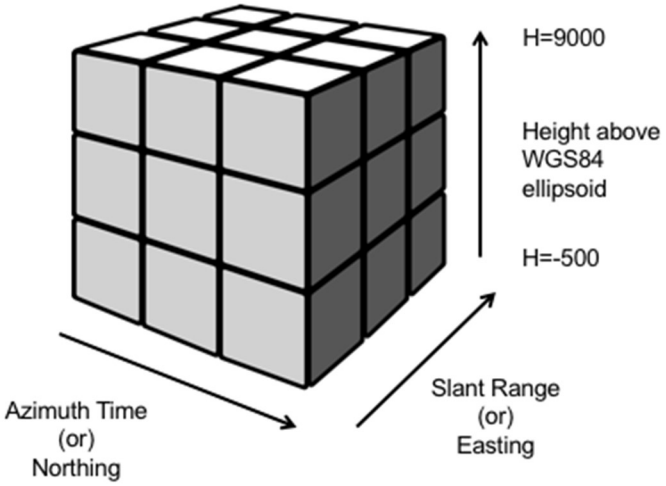


Figure 6-1. Metadata cube layer schematic

6.1 Metadata Cube Interpolation Example

We provide here a conceptual example of how these metadata cubes can be used within an existing GIS framework. Let us consider a GUNW product on a UTM Zone 10 grid. We use a

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geocoded product for the demonstration, but the presented approach can be easily extended to radar coordinate products by replacing northing axis by azimuth time and easting axis by slant range.

Table 6-1 Example metadata cube properties.

Name	Value	Description
Primary layer properties		
xmin	100000.0	Easting of the first column (m)
xmax	340000.0	Easting of the last column (m)
dx	30.0	Column spacing in Easting (m)
Nx	8001	Number of columns
ymax	570000.0	Northing of first row (m)
ymin	330000.0	Northing of last row (m)
dy	-30.0	Row spacing in Northing (m). Negative to emphasize North-up imagery in geocoded products
Ny	8001	Number of rows
Metadata cube properties		
Cxmin	97000.0	Easting of first column (m)
Cxmax	343000.0	Easting of last column (m)
Cdx	1000.0	Column spacing in Easting (m)
CNx	247	Number of columns
Cymax	579000.0	Northing of first row (m)
Cymin	321000.0	Northing of last row(m)
Cdy	-3000.0	Row spacing in Northing (m). Negative to emphasize North-up imagery in geocoded products
CNy	87	Number of rows
Czmin	-1500	Height of the first layer (m)
Czmax	9000	Height of the last layer (m)
Cdz	1500	Layer spacing in height (m)
CNz	8	Number of height layers

Suppose we are interested in computing the Perpendicular Baseline (Bperp) at a pixel of interest located at UTM coordinates point (Px,Py). Since these are coordinates on a map domain, we can look up a DEM to get the height at this point. The three-dimensional point of interest then becomes (Px, Py, h(Px,Py)).

The metadata cube for Perpendicular baseline can be thought of as a three-dimensional field Bperp(x,y,z) – even though it is oriented as (Nz,Ny,Nx) in the HDF5 file for ease of use with a GIS. The user can use standard built-in regular grid three-dimensional interpolation routines in languages like MATLAB (e.g, interp3), IDL or Python (e.g, RegularGridInterpolator) to interpolate the Bperp array. We recommend cubic interpolation for best results. If a three-dimensional interpolator is not available, one could use two-dimensional cubic interpolation for each height layer followed by a one-dimensional cubic interpolation in the following manner:

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1. Populate f(i), i=0,…Nz-1 by two-dimensional cubic interpolation of each height layer:

$$f(i) = Bperp\left[i, \frac{Py - Cymax}{Cdy}, \frac{Px - Cxmax}{Cdx}\right]$$

where the numbers in the square brackets indicate indices into the three-dimensional cube. For example, if we are interested in the point (107590.0 East, 555870.0 North, 300.0 Height), we would interpolate at Row 7.71 and Column 10.59 for each height layer.

2. Interpolate f(i) using one-dimensional cubic interpolation:

$$Bperp\big(Px, Py, h(Px, Py)\big) = f\left[\frac{h(Px, Py) - Czmin}{Cdz}\right]$$

where the number in the square bracket indicates an index into a one-dimensional array. For example, for a height value of 200.0, we would interpolate at an index of 1.2.

6.2 Metadata Cube Usage Note

Note that the metadata cubes are designed to accommodate one double-precision cube within 1 MB of memory, allowing for information to be easily stored in memory for on-the-fly computation within GIS frameworks or software without much overhead. The metadata cubes are not a replacement for traditional SAR processing approaches or very high-resolution analyses. They are meant to facilitate rapid processing and analysis by non-experts and will serve the needs for most SAR applications. Analyses show that the geolocation error is on the order of 1.5 cm due to interpolation which is significantly smaller than errors from sources such as DEM, orbits, and atmospheric path delay. Interpolation errors for each of the metadata layers will be reported after additional study.

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7 APPENDIX A: ACRONYMS

ADT	Algorithm Development Team
AT	Along Track
AWS	Amazon Web Services
BFPQ	Block adaptive Floating-Point Quantization
Cal/Val	Calibration and Validation (also sometimes cal/val)
CDR	Critical Design Review
CF	Climate and Forecast
CPU	Central Processing Unit
CRSD	Calibration Raw Signal Data
CSV	Comma-separated values
DAAC	Distributed Active Archive Center
DEM	Digital Elevation Model
DN	Digital Number
EAR	Export Administration Regulations
ECMWF	European Centre for Medium-Range Weather Forecasts
ECEF	Earth Centered Earth Fixed
EPSG	European Petroleum Survey Group
ESA	European Space Agency
FM	Frequency Modulation
FOP	Forecast Orbit Ephemeris
FOV	Field of View
GCOV	Geocoded Polarimetric Covariance (also as L2_GCOV)
GCP	Ground Control Point
GDAL	Geospatial Data Abstraction Library
GDS	Ground Data System
GIS	Geographic Information System
GMTED	Global Multi-resolution Terrain Elevation Data
GOFF	Geocoded Pixel Offsets (also as L2_GOFF)
GPU	Graphics Processing Unit
GSLC	Geocoded Single Look Complex (also as L2_GSLC)
GUNW	Geocoded Unwrapped Interferogram (also as L2_GUNW)
HDF5	Hierarchical Data Format version 5
HK, HKTM	Housekeeping Telemetry
InSAR	Interferometric Synthetic Aperture Radar
ISCE	InSAR Scientific Computing Environment
ISCE3	InSAR Scientific Computing Environment Enhanced Edition (for NISAR)
ISO	International Organization for Standardization
ISRO	Indian Space Research Organisation (British spelling)
L0B	Level-0B (data)

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L1	Level-1 (data)
L2	Level-2 (data)
LOS	Line-Of-Sight
LUT	Lookup Table
Mbps	Megabits per second
MHz	Megahertz
MOE	Medium-precision Orbit Ephemeris
NCSA	National Center for Supercomputing Applications
NetCDF4	Network Common Data Form version 4
NISAR	NASA-ISRO Synthetic Aperture Radar
NOE	Near-Realtime Orbit Ephemeris
PDR	Preliminary Design Review
PLM	Product Lifecycle Management
POD	Precision Orbit Determination
POE	Precision Orbit Ephemeris
PRF	Pulse Repetition Frequency
QA	Quality Assurance
REE	Radar Echo Emulator
RFI	Radio Frequency Interference
RIFG	Range-Doppler Interferogram (also as L1_RIFG)
ROFF	Range-Doppler Pixel Offsets (also as L1_ROFF)
RRSD	Radar Raw Signal Data
RRST	Radar Raw Science Telemetry
RSLC	Range-Doppler Single Look Complex (also as L1_RSLC)
RUNW	Range-Doppler UnWrapped Interferogram (also as L1_RUNW)
SAR	Synthetic Aperture Radar
SAS	Science Algorithm Software
SDS	Science Data System
SDT	Science Definition Team
SIS	Software Interface Specification
SLC	Single Look Complex
SNAPHU	Statistical-cost, Network-flow Algorithm for Phase Unwrapping
SRTM	Shuttle Radar Topography Mission
ST	Science Team
TAI	International Atomic Time (Temps Atomique International)
TCF	Terrain Correction Factor
TEC	Total Electron Content
TFdb	Track-frame Database
SWST	Sampling Window Start Time
UR	Urgent Response
UTC	Universal Time Coordinated

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UTM	Universal Transverse Mercator
WGS84	World Geodetic System 84
XML	eXtensible Markup Language (xml in code)
YAML	YAML Ain’t Markup Language

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8 APPENDIX B: GEOCODED PRODUCT GRIDS

NISAR L2 products will be generated on a pre-defined Track/Frame system. The projection system for a particular frame will be available to the users as a predefined map and will be held constant through the life of the system. Each L2 HDF5 granule itself will include information indicating the projection used for the product.

Map Projections

NISAR's SDS is able to ingest any DEM whose vertical datum represents height above the WGS84 ellipsoid and the horizontal datum can be represented by an EPSG code for generating geocoded product. Table 7-1 lists the various projection systems used to output L2 geocoded products.

Table 8-1 Projection systems for NISAR L2 products.

EPSG code	PROJ.4 string	Common Name	Geographical scope
3031	+proj=stere +lat_0=-90 +lat_ts=-71 +lon_0=0 +k=1 +x_0=0 +y_0=0 +datum=WGS84 +units=m +no_defs	Antarctic Polar Stereographic	Antarctica and Southern Hemisphere Sea Ice
3413	+proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +k=1 +x_0=0 +y_0=0 +datum=WGS84 +units=m +no_defs	NSIDC Sea Ice Polar Stereographic North	Greenland and Northern Hemisphere Sea Ice
32601-32660	+proj=utm +zone=X-32600 +datum=WGS84 +units=m +no_defs	UTM Zone North	Northern Hemisphere Land except Greenland
32701-32760	+proj=utm +zone=X-32700 +south +datum=WGS84 +units=m +no_defs	UTM Zone South	Southern Hemisphere Land except Antarctica

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